

TOR1: Climate & Environmental Considerations

An exploration of thermal habitat changes and risk in
the Mid-Atlantic Bight using GLORYS, 1993-2023

Joseph Caracappa, Deborah Hart, Jui-Han Chang, Kimberly Hyde

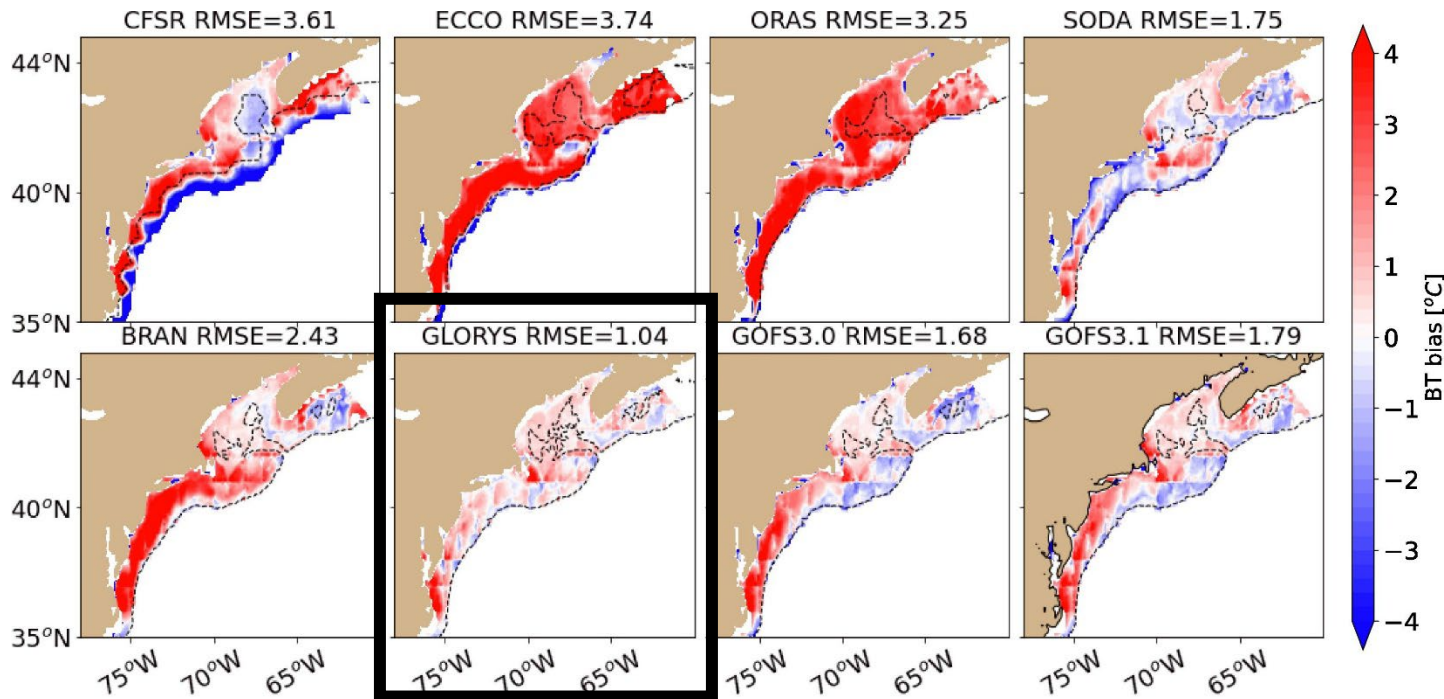
Overview

Mortality events in the Mid-Atlantic Bight initiated a review of potential heat stress events impacting scallop populations

Objectives

- Characterize thermal habitat for sea scallops
- Develop thermal stress risk indicators
- Identify changes over time

GLORYS Reanalysis



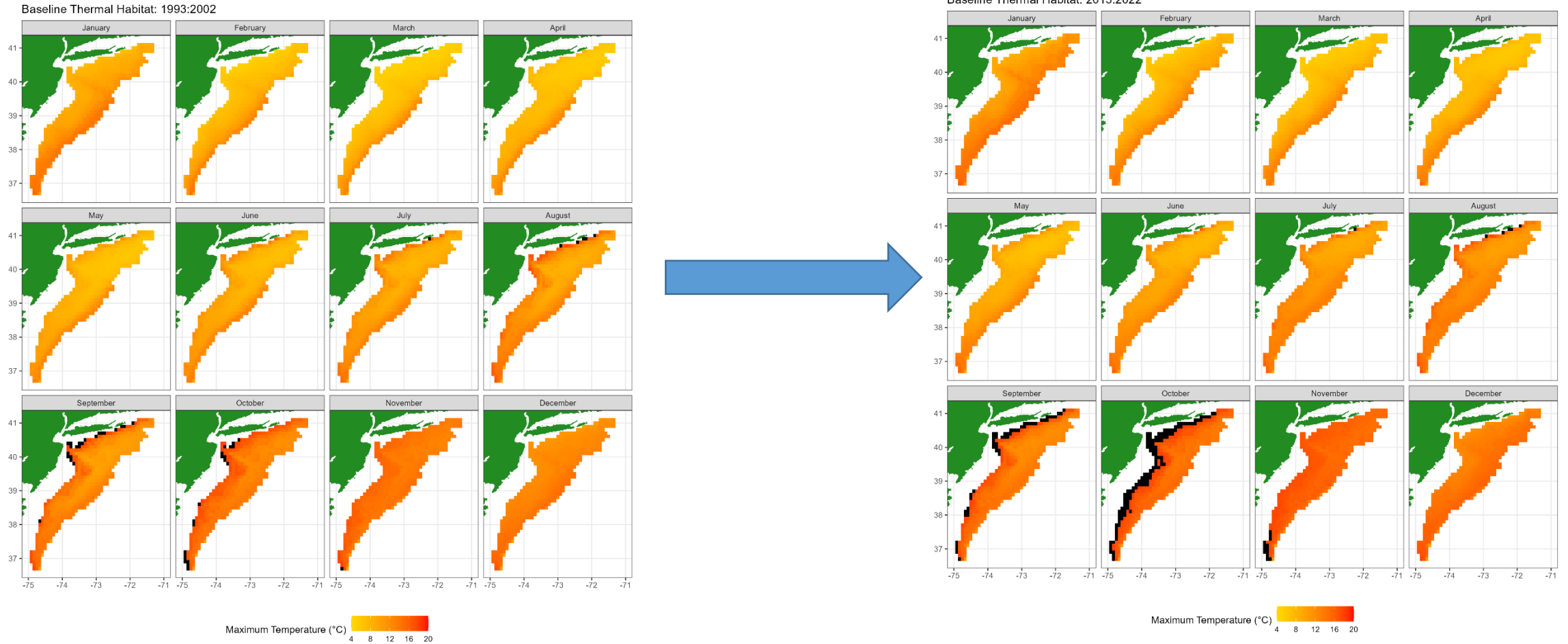
Using GLORYS reanalysis for bottom temperature

- A global product from Copernicus
- 1/12° resolution
- Daily from 1993 – present
- Top performing reanalysis product for bottom temperature in northeast US

Thermal Habitat Characterization & Heat Stress Event Definition

- Heat Stress Events: Locations where temperature exceeds 18°C for an extended period of time
- Thermal habitat
 - **Baseline thermal habitat:** Spatial area where *maximum* monthly bottom temperature *never* exceeds 18°C.
 - Describes where thermal conditions are generally suitable for scallops
 - **Unsuitable thermal habitat:** Spatial area where *mean* monthly bottom temperature exceeds 18°C
 - Describes where thermal conditions are *not* suitable for scallops

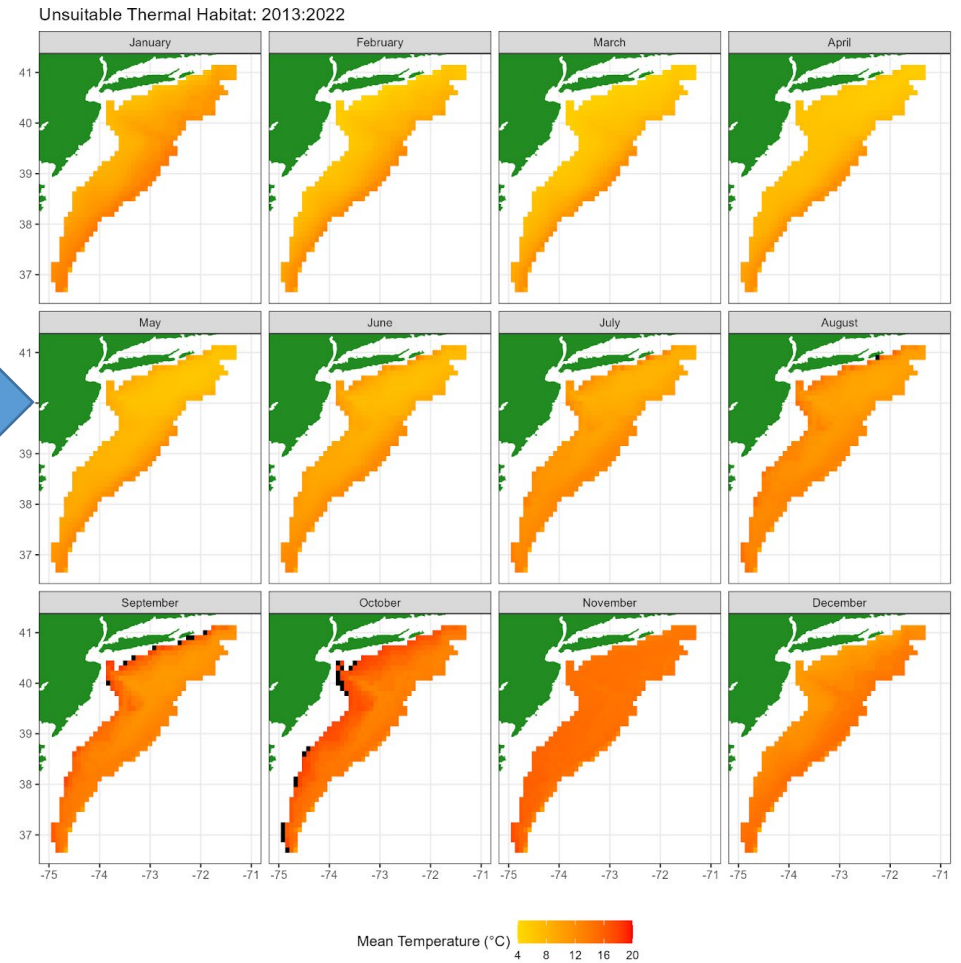
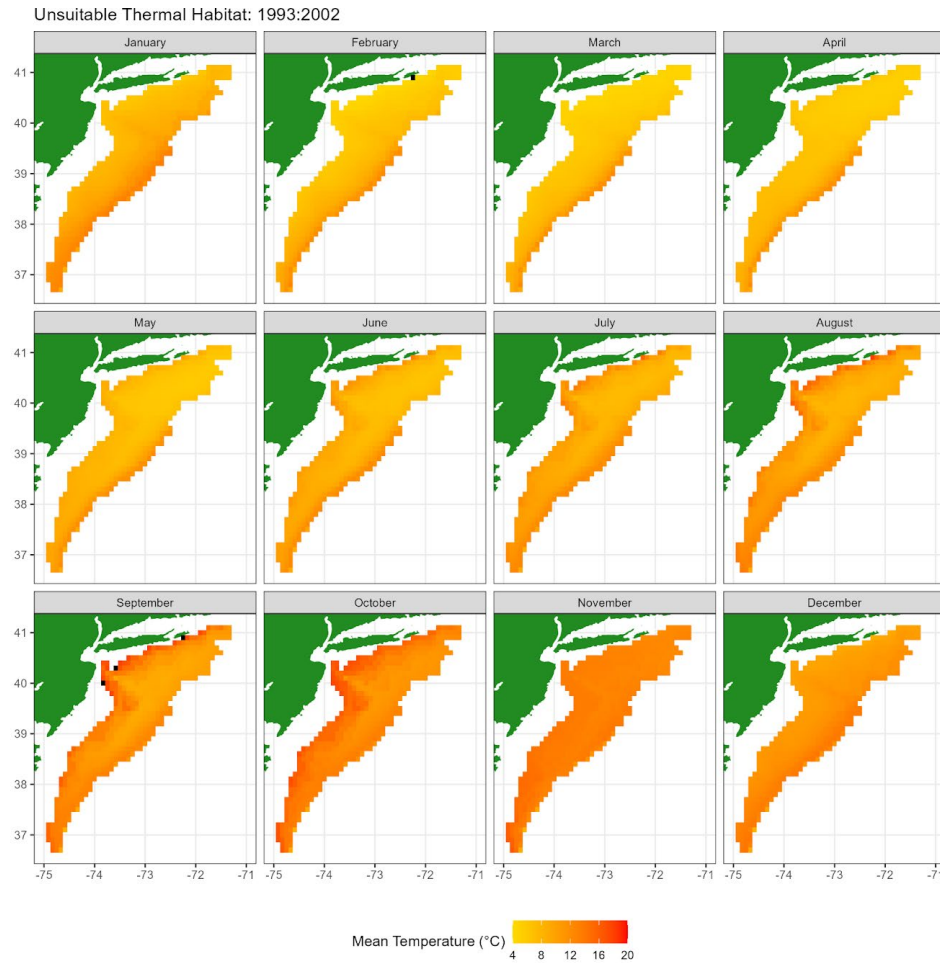
Baseline Thermal Habitat



Baseline thermal habitat defined as areas where maximum monthly bottom temperature always $< 18^{\circ}\text{C}$ between 25 and 100m

Large decrease in thermal habitat in recent decade within inshore and southern MAB.

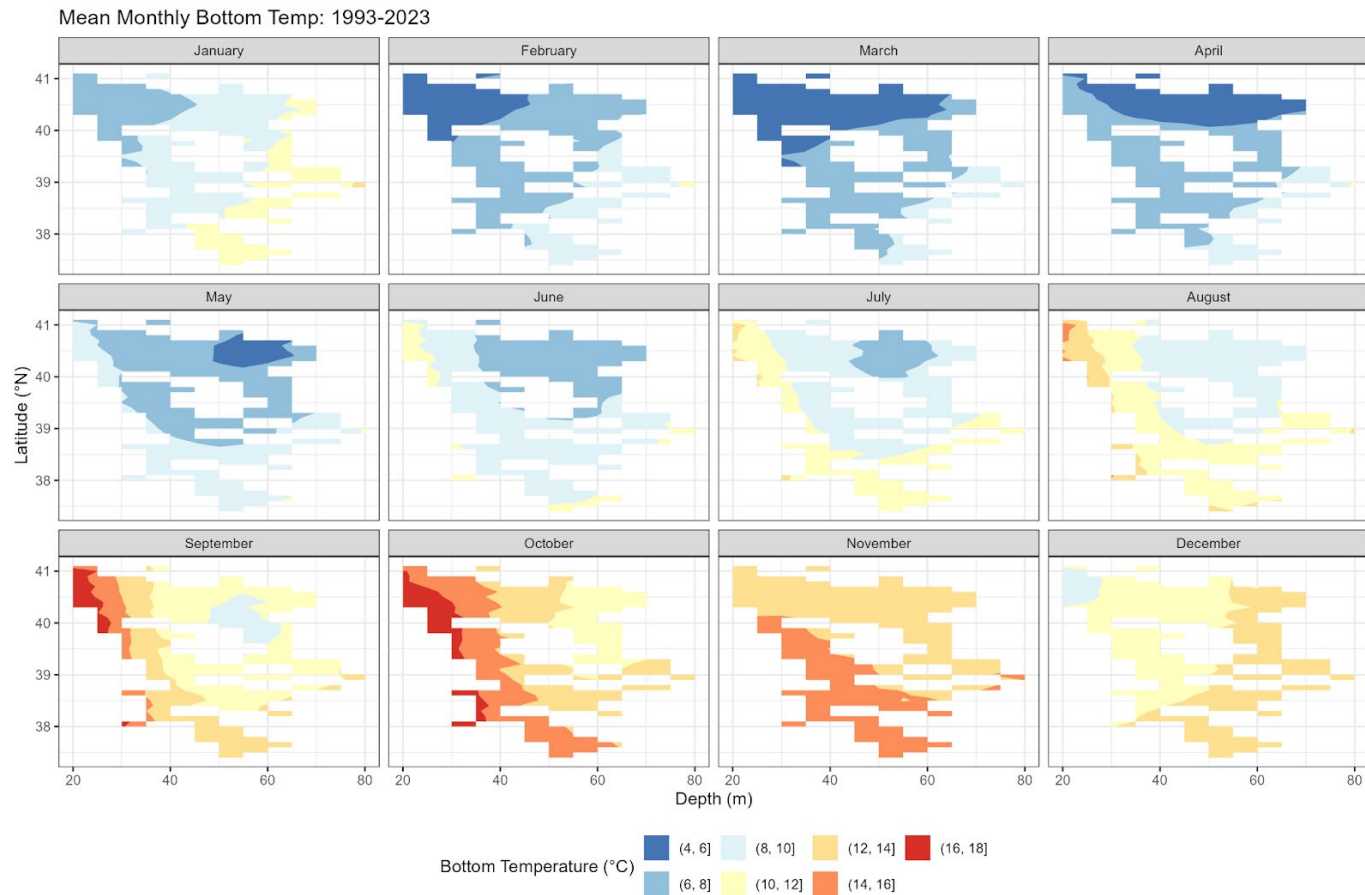
Unsuitable Thermal Habitat



Not much unsuitable thermal habitat

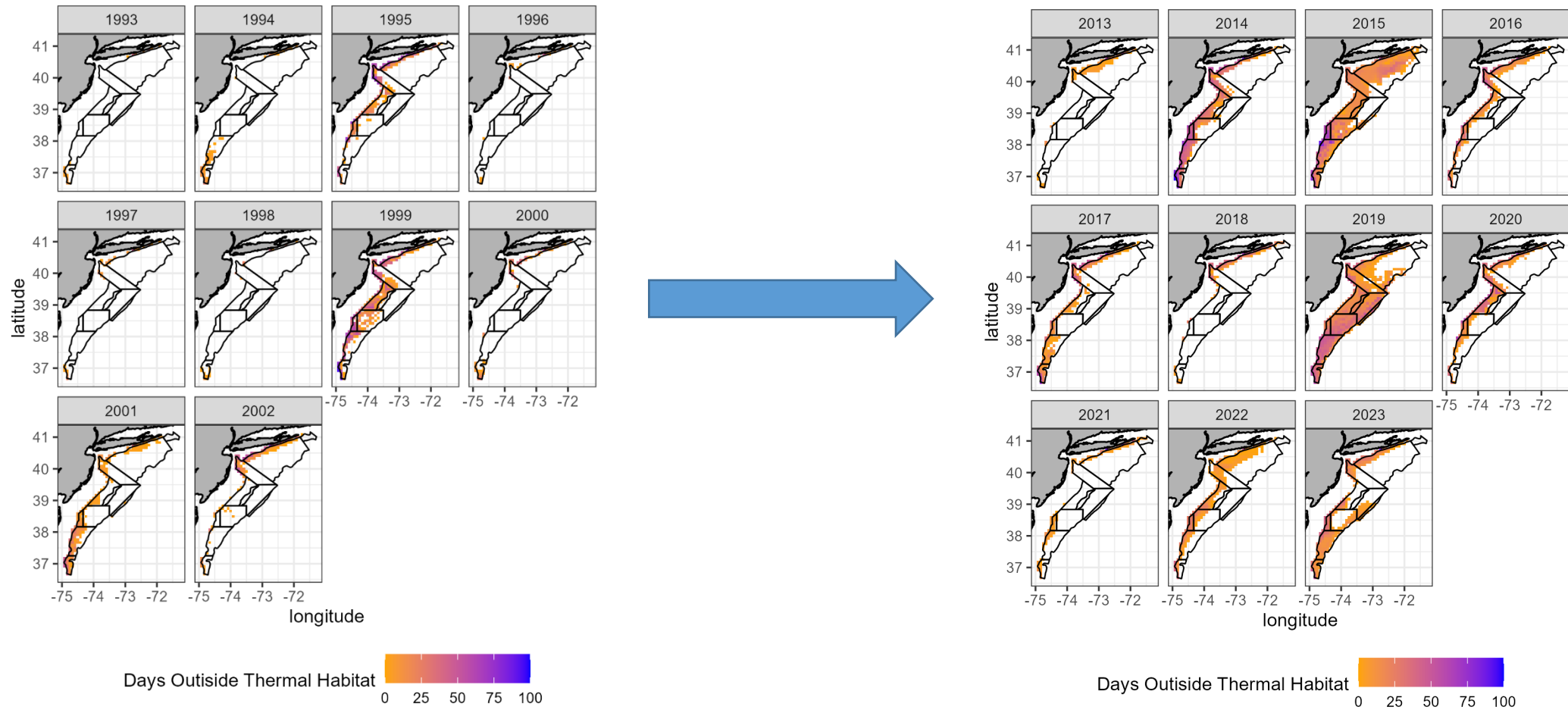
Increases in September-October in inshore MAB in recent decade

Unsuitable thermal habitat



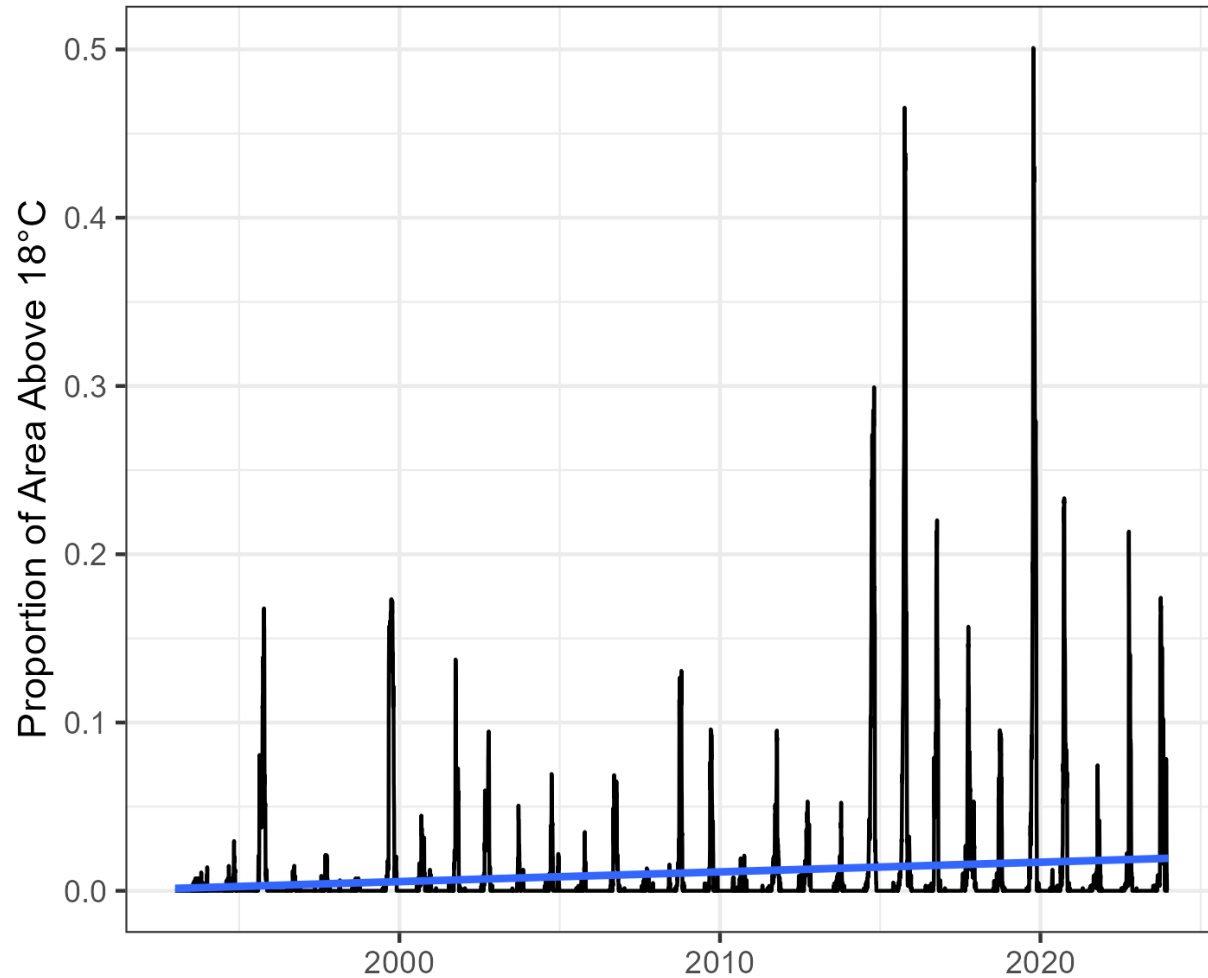
- Over entire GLORYS duration (1993-2023), no areas with scallop management area exceed a mean monthly bottom temperature of 18°C.
- These conditions aren't "normal" in any part of the study region
- Recent heat stress events are anomalous, but becoming more regular.

Duration of Heat Stress Events



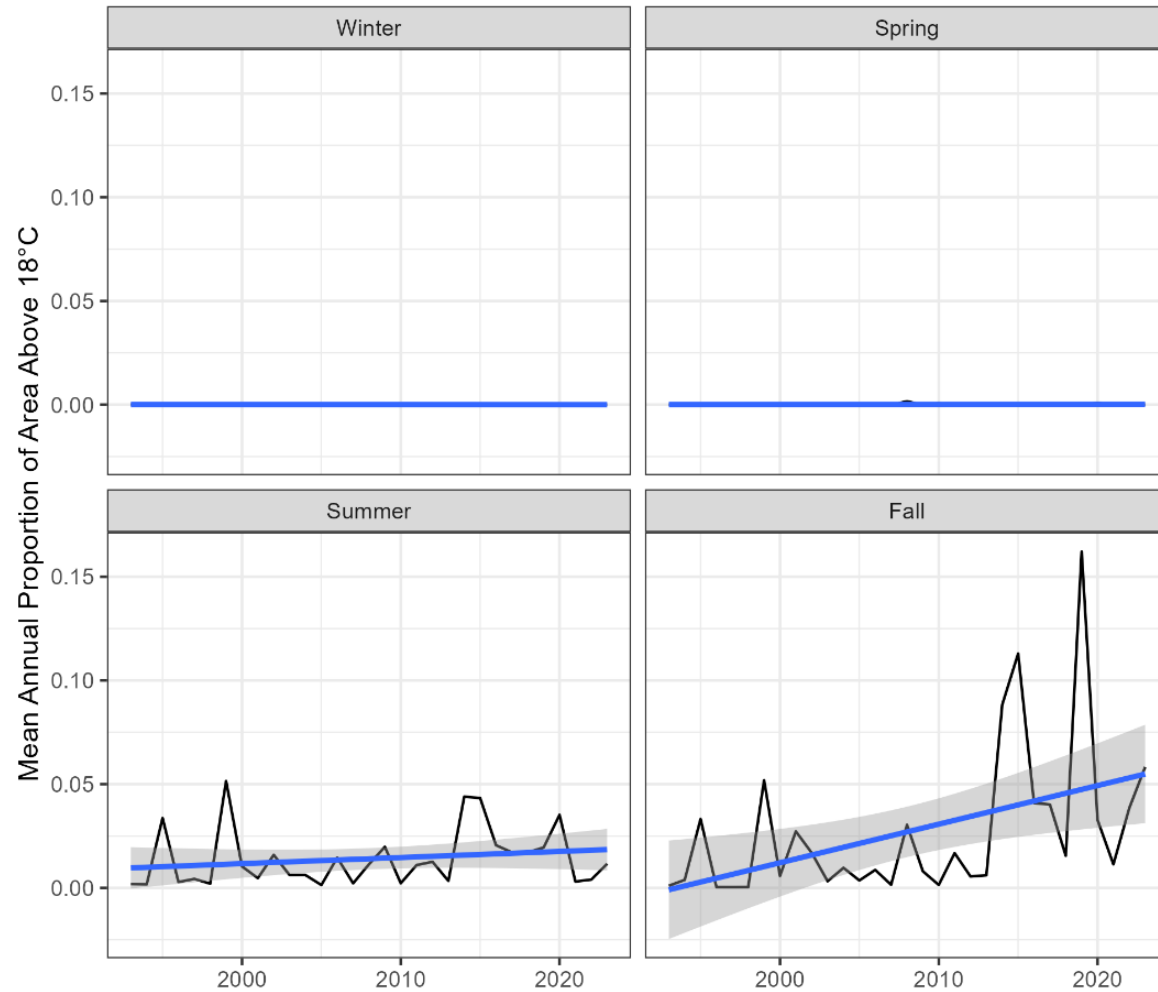
Heat stress events when bottom temperature $> 18^{\circ}\text{C}$ and can occur from days to weeks
Large increase in heat stress events in recent decade (especially 2015 and 2019)

Extent of Heat Stress Events



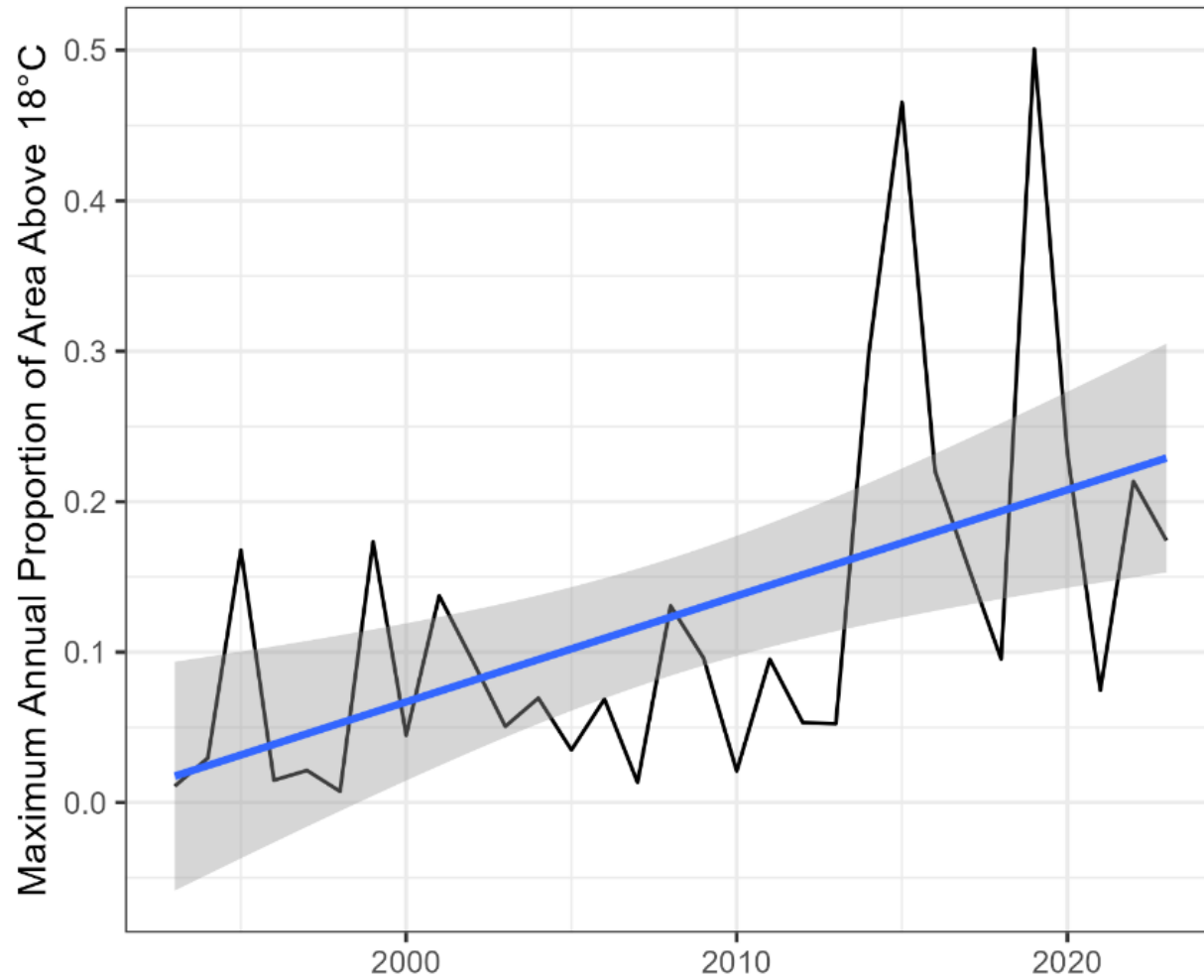
- Proportion of scallop management areas that experience heat stress events by day
- Large events in recent decade in fall
- Increasing trend in the extent of events
- Peak daily extent in last 10 years, 7-50% of managed area

Extent of Heat Stress Events



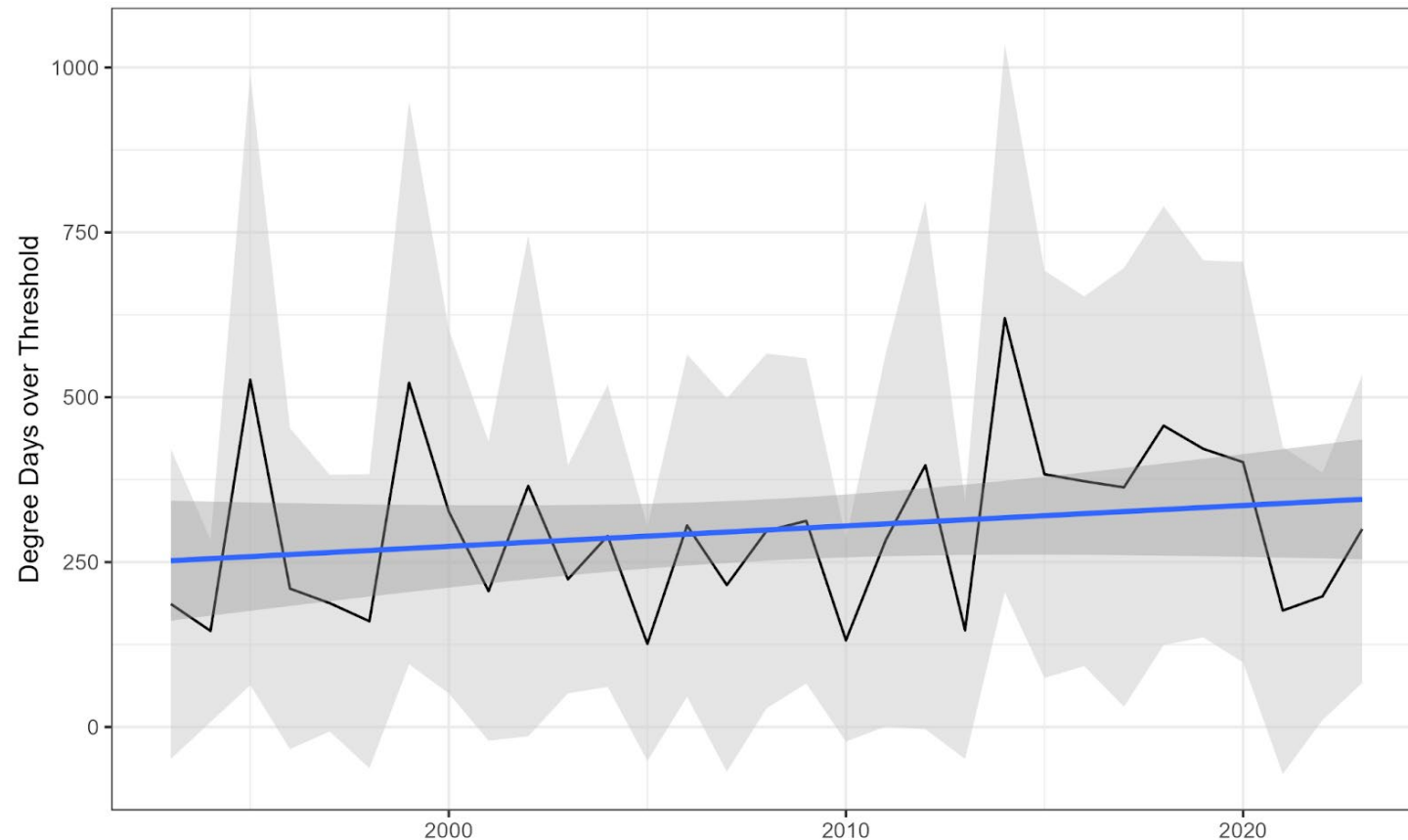
- Heat stress events mostly occur in summer (June-August) and Fall (Sept-Dec.)
- Extent of fall events have been increasing

Extent of Heat Stress Events



- Extent of largest annual heat stress event has been increasing

Heat stress event intensity



- Intensity measured as degree days exceeding 18°C
- Hypothesis is that more intense heat stress events would result in higher mortality in shorter time.
- No significant trend ($R^2 = 0.015$, $p = 0.23$)
- Despite being more prevalent and widespread, not getting more intense.

Conclusions

1. Baseline thermal habitat for sea scallops is contracting in the MAB
2. There is an increase in the extent and duration of heat stress events, especially in September and October
3. Heat stress doesn't fully explain observed mortality events. Other possible factors include:
 - Food availability
 - Bottom oxygen
 - Amount of scallop energy reserves (e.g. glycogen)